MTH212M PRACTICE PROBLEMS WEEK 1

Question 1. Fix $p \in (0,1)$ and let $\{X_n\}_{n\geq 1}$ be an i.i.d. sequence of *Bernoulli*(p) RVs. Define $Y_n := \min\{10, X_1 + X_2 + \cdots + X_n\}, n = 1, 2, \cdots$. Show that $\{Y_n\}_{n\geq 1}$ is a Markov Chain. Also, identify the state space and the transition probability matrix.

Question 2. Let P be a stochastic matrix. Is it true that $P^n, n = 2, 3, \cdots$ are also stochastic matrices?

Question 3. Recall the two person's Zero-sum game discussed in the class. Assume that initially the two persons A and B start with 10 Rupees and 5 Rupees respectively. Let X_n denote the amount of money held by A after the n-th toss. If a fair coin is used in the tosses, then find the following:

(i) the state space S

(ii) the transition probability matrix P

(iii) the initial distribution, i.e. the law of X_0

Question 4. A urn contains b black balls and w white balls. Draw a ball at random, and replace it by a ball of the other colour. Let X_n denote the number of black balls after the n-th draw. Show that $\{X_n\}_{n\geq 0}$ forms a Markov Chain. Also find the state space and the transition probability matrix.

Question 5. Consider a Markov Chain $\{X_n\}_{n=0}^{\infty}$ with time-dependent one-step transition probabilities $p_{ij}^{(n)} := \mathbb{P}(X_{n+1} = j | X_n = i)$. Define a sequence $\{Y_n\}_{n=0}^{\infty}$ given by $Y_n := (n, X_n)$.

(i) Identify the state space of Y_n's in terms of that of X_n's.

(ii) Show that $\{Y_n\}_{n=0}^{\infty}$ form a Markov Chain with stationary transition probabilities.